



Course Code: IS364TA

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(An Autonomous Institution Affiliated to VTU)
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Mysuru Road, Bengaluru - 560059

(Common to CD, CS, CY, IS)

Software Engineering with Agile Technologies

Maximum Marks : 100

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Question No	Question	M	CO	BT
1.1	In an Agile project, which Scrum artifact contains the list of features to be developed?	01	4	2
1.2	How does adherence to ethical standards contribute to software quality and user trust?	01	4	4
1.3	Which Agile methodology uses fixed-length iterations called sprints?	01	4	1
1.4	In Agile development, which artifact is commonly used to represent requirements?	01	4	3
1.5	The UML diagram used to represent Structural Model is _____ and Behavioural Model is _____.	02	4	2
1.6	Highlight the importance of UML diagrams and how does it help in implementation?	02	3	2
1.7	Identify one situation that makes object class testing more complicated	01	3	3
1.8	Requirements-based testing is validation rather than defect testing. Justify the statement	02	3	4
1.9	The process of making improvements to a program to slow down degradation through change is called _____	01	3	1
1.10	Which are the three stages in a project life cycle where project planning takes place.	02	1	2
1.11	List any two characteristics of Plan-Driven Development.	02	1	1
1.12	Define the term "velocity" and "potentially shippable product increment" with respect to Scrum context	02	4	1
1.13	Define the role of an Agile Coach.	02	4	1

Part B

Question

Question No

Identify whether the statements below map to Functional or Non functional requirements.

M CO ET

2a	i. The system shall ensure only authorized users can access data.			
	ii. The system shall allow patients to book, reschedule, and cancel appointments.			
	iii. The system shall manage room/bed allocation for admitted patients.			
	iv. The system shall allow airport staff to update flight status (boarding, delayed, cancelled).			
2b	Differentiate between Plan driven and Agile process with examples.	06	4	4
2c	Differentiate between Functional and Non Functional Requirements with examples.	06	3	4
3a	Briefly explain the Repository architectural pattern with a suitable example along with its advantages and disadvantages.	08	4	2
3b	A web browser sends requests to centralized servers hosting websites and receives responses. Which architecture is suitable? Justify your answer with advantages and disadvantages.	08	4	3

OR

	Explain the importance of Software reuse at the following levels:			
4a	(i) Abstraction level (ii) Object level (iii) Component level (iv) System level	08	3	3
4b	Explain the concept of Flow in Kanban. Discuss the various metrics used to measure and manage flow, and explain how these metrics help improve process efficiency and delivery performance.	08	4	2
5a	Explain the different types of user testing. With the help of a neat diagram, explain the acceptance testing process	08	3	2
5b	Using a neat diagram, explain CBSE with reuse. Bring out the essential differences between CBSE with reuse and software processes for original software development	08	2	2

OR

6a	Illustrate and explain the general model of the software evolution process	08	1	2
6b	What is release testing? Explain requirements-based testing and scenario testing as approaches used in release testing	08	3	2
7a	Illustrate the factors that influence the price quoted for a software product.	08	4	3
7b	How does agile planning handle changes and risks during development? Explain with an example.	08	4	4

8a	Describe Software Project Planning and its major activities.	08	2	2
8b	Discuss staffing patterns and schedule constraints in software projects. Explain why adding more people may delay a project.	08	2	4
9a	For large, long-lifetime systems that are developed by a software company for an external client, using an agile approach, presents several problems. Identify and explain a few. Also identify the key challenges in using Agile	08	4	3
9b	Explain the Principles of Coaching in Agile development. Discuss how these principles help teams become self-organizing and continuously improving.	08	4	3

OR

10a	Explain the factors that contribute to the successful adoption and implementation of a methodology in an organization. What role does an Agile Coach play in ensuring its effectiveness?	08	4	4
10b	A startup develops an online food delivery application. Initially, customers requested only food ordering. During development, they requested live order tracking, online payment, coupons, and restaurant ratings. The company wants to accommodate these changes without restarting the project.	08	4	4
	Explain how Agile Software Development can help manage changing requirements. Discuss the Agile principles that support this scenario.			